Interview with SHRI. N. S. BOSERAJU, Minister for Minor Irrigation, Science & Technology, Karnataka.

1. Why was the Quantum India Summit 2025 significant for Karnataka and India?
The Quantum India Summit 2025 is the first of its kind summit in India. Hosting over 1,950 participants from 19 countries far exceeding the expected 600–700 the event brought together Nobel Laureates, Consulate Generals, scientists, industry leaders, and startups under one roof. We used this platform to showcase our Quantum Vision and Roadmap to the world which included the formation of a Quantum Task Force and the announcement of a dedicated Quantum City. The summit has positioned Karnataka on the forefront of India's global quantum journey.

2. What is the role of the Karnataka Quantum Task Force?
The Karnataka Quantum Task Force will function as the apex advisory and coordination body guiding the implementation of our quantum roadmap. It will include experts from academia, government and industry. This body will be pivotal in building a complete supply chain from R&D and talent to manufacturing and global partnerships, ensuring Karnataka doesn't just adopt quantum tech, but exports it.

3. How does Karnataka's approach differ from other state initiatives like Amaravati?
Bengaluru is the ideal location for India's quantum ecosystem because it already hosts a thriving quantum base along with robust R&D Infrastructure. We are home to India's first commercially operating quantum computer which was built indigenously and a quantum chip manufacturing facility is starting shortly. While some states focus primarily on data centers or digital warehousing, we are focussed on building a holistic deep-tech ecosystem from fundamental research to full-stack manufacturing. Karnataka is not competing with other Indian states we are benchmarking against global leaders.

4. What is the idea behind Quantum City and how will it enable industry development?
Quantum City is envisioned as an integrated hub that will bring together R&D, component manufacturing, system integration, skilling, and incubation. The government plans to provide infrastructure for Hardware park, plug-and-play incubation support, and incentives to foster startups. Our strategy is to build an ecosystem where the entire quantum value chain coexists from labs to fabs.

5. Isn't $20 billion quantum economy vision by 2035 too ambitious ?
We believe governments must dream big and build structured pathways to achieve ambitious goals. After the success of the Quantum India Summit we are confident that this vision is not only feasible but necessary. Globally, the quantum market is projected to exceed $150 billion by 2035. We aim to capture a meaningful share by becoming an innovation and export hub.

6. What were the most valuable takeaways from the roundtable conference at the summit?
The roundtable chaired by Hon'ble DCM Shri DK Shivakumar brought together Industry leaders, academic researchers, startup founders, policymakers, Consul Generals and representatives from trade councils. This meet provided key insights into India's current Quantum gaps, particularly in manufacturing, IP creation, start-up ecosystem and skill development. This meet has helped us to understand the shortcomings and build our Quantum Stragey to solve the current infrastructure and academic gaps and create a more product-based approach. A strong aspiration to achieve indigenous manufacturing capabilities has emerged. We must move fast and this should be the priority going forward. Several countries expressed their willingness in cross-country collaborations. Looking at the positive response, we are planning to take up roadshows across the world and conduct the summit annually.

7. Nobel Laureates at the summit said India has talent, but lacks science funding and opportunities. How does Karnataka respond to this concern?
We deeply value this observation by the Nobel Laureates; it's a reminder that nurturing talent must go hand-in-hand with providing institutional and financial support. The Union funding for R&D compared to the GDP is not sufficient, but Karnataka has already taken concrete steps to address this gap. We've sanctioned India's first dedicated Quantum Research Park at IISc and plan to establish a Quantum City with a dedicated Quantum Hardware Park. Honorable Chief Minister has announced ₹ 1,000 crores which will be dedicated towards developing this Infrastructure and also includes an innovation fund which will support at least 100 startups and deliver 100 patents with the goal of developing 1000 qbit processing power indigenously. We are developing academic quantum nodes across regional universities and incorporating activity-based quantum curriculum at the high school level. An MoU has been already been signed between IIIT Dharwad and QpiAI to host an academic quantum computer. Most importantly, we are aligning our education and skilling policies to empower youth in this space. We have learnt our lessons from the Semiconductor industry, where India lagged in leveraging it, Karnataka is committed to not repeating the same mistakes with quantum technology.

education will be introduced at the senior school level in both **English and Kannada**. Additionally, the **DST PhD fellowship program** will be expanded to train a new generation of quantum scientists.

A Catalyst for Startups and Industry

The state is backing quantum startups with a **Quantum Innovation Fund** and venture capital support. With plans to nurture **100+ startups** and facilitate **100+ patent filings,** Karnataka is building an ecosystem that encourages risk-taking and innovation. Several private players, including **IBM, AWS, Infosys, XeedQ, and L&T,** showcased their quantum initiatives at the summit, further cementing industry-government collaboration.

Why Quantum Matters

Quantum technology promises breakthroughs that can redefine industries. From early **disease detection to secure communication, smarter agriculture,** and **national security,** its applications are vast. Globally, countries are racing to dominate this field, and Karnataka is positioning itself as India's spearhead.

Looking Ahead

The success of the **Quantum India Bengaluru Summit 2025** is only the beginning. The state has announced plans to **host the summit annually**, making it a recurring platform for global thought leadership. Roadshows across India and abroad will further showcase Karnataka's readiness to lead the quantum revolution.

As Minister Boseraju aptly concluded, "Quantum is not just technology—it's about inclusive progress, dignity, and development. With our roadmap, Karnataka will not only lead India but shape the future of quantum innovation globally."

The Road to 2035

With its **clear roadmap, funding support,** and **ecosystem advantage,** Karnataka is on course to achieve its **Quantum Vision 2035.** If successful, the state will not only capture **20% of the global quantum market but** also create **over two lakh direct jobs,** making India a formidable player in the next technological revolution.

Global Roundtable Drives Collaborative Quantum Future:

A key highlight of the Quantum India Bengaluru 2025 Summit was the high-level roundtable meeting chaired by Deputy Chief Minister D.K. Shivakumar. The session brought together representatives from 19 countries, including the USA, UK, Germany, Japan, and Israel, along with leading global organizations in the quantum sector. Participants discussed strategic collaborations, investment opportunities, and policy frameworks to accelerate Karnataka's quantum growth. Several nations expressed interest in forming joint ventures and setting up research centers in the state. The discussions underscored Karnataka's readiness to serve as a global partner in shaping the future of quantum technologies.



Hon'ble Minister for Minor Irrigation, Science & Technology Shri N.S. Boseraju delivering his address at the inaugural session of the Quantum India Bengaluru Summit 2025.